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Healthcare Innovation

An exploratory multi-scale analysis linking MRI radiomics with KDR/VEGFR-2 expression and kinase-domain structure in glioblastoma

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CMU
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Glioblastoma

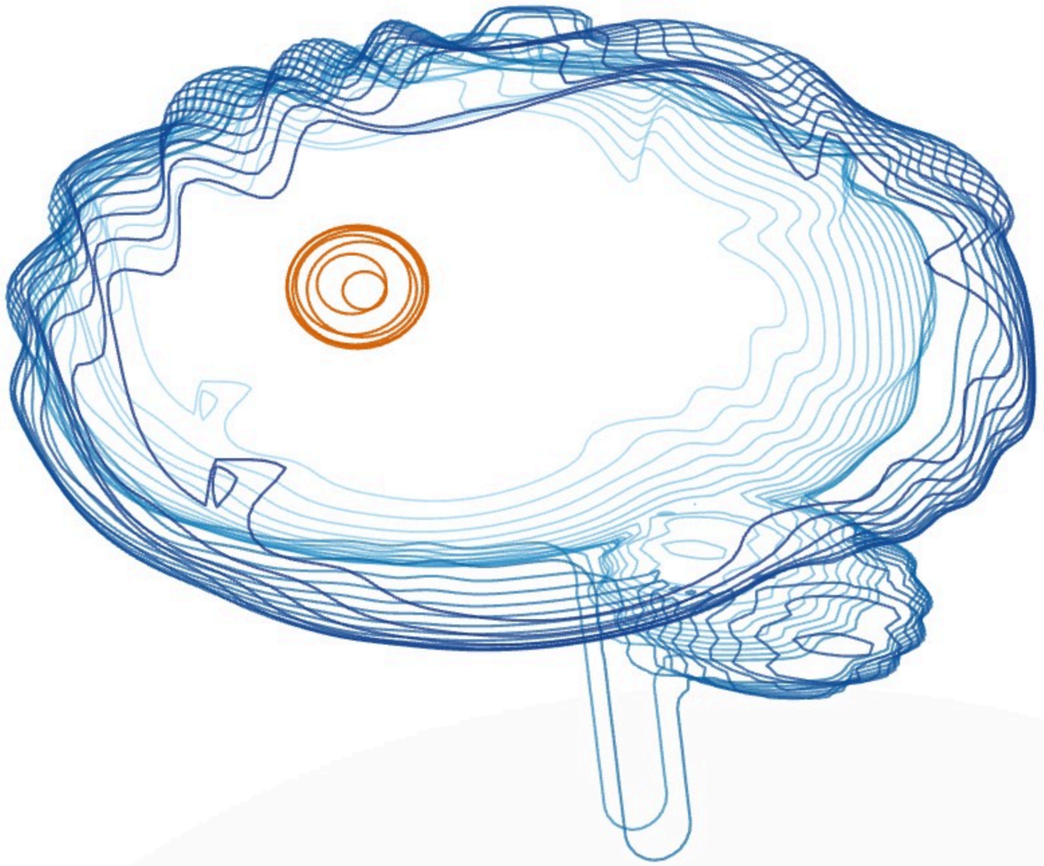
15 months

median survival, glioblastoma

That is how long most people live after they are told they have this brain cancer. It is called glioblastoma, and it is the most dangerous one we know.

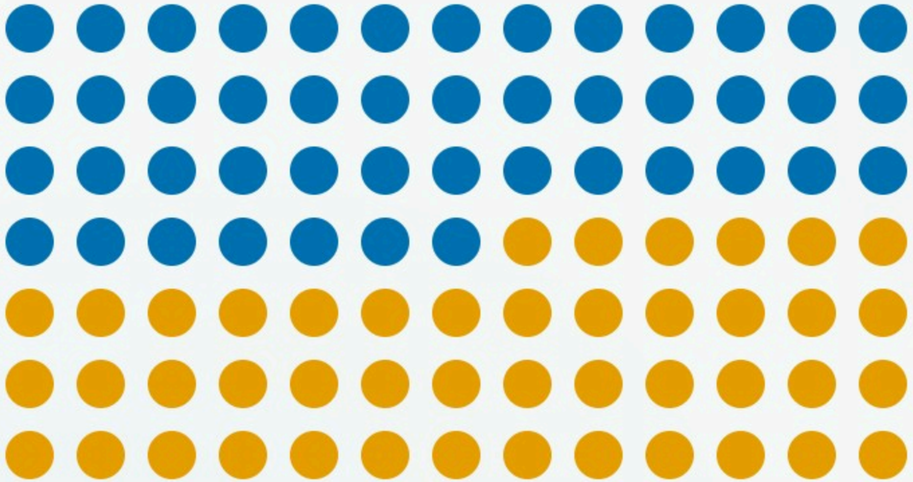
There is one drug that helps some of these patients. For the rest, it does nothing.

And today, the only way to find out which group a patient is in is surgery.



Background: illustrative animation, not data.

KDR expression · median split



46 HIGH

45 LOW

The question

Measure KDR from an MRI, not from tissue.

Bevacizumab	Blocks the signal that grows new blood vessels into the tumour.
VEGFR-2 · KDR	The receiver, and the gene that builds it.

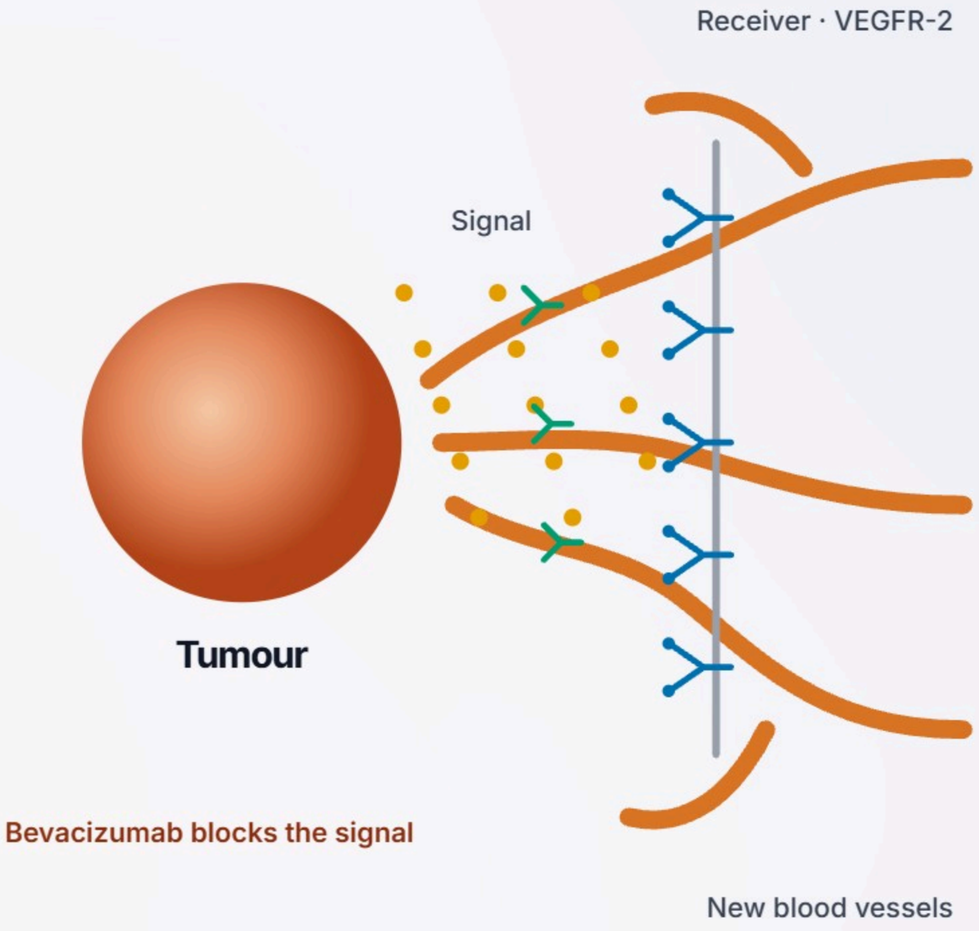
91 patients · all public data

No new scans, no new procedures, nothing a hospital does not already have.

Background

A signal does nothing unless something receives it.

That is why KDR is our target, and why reading it without surgery would be useful.



01 A tumour grows faster than its own blood supply.

02 It sends a signal for new blood vessels: angiogenesis.

03 More receivers means more blood vessels, and more for the drug to block.

Image
MRI · T1, T1-Gd, T2, FLAIR

704 pre-computed · 1,098 PyRadiomics descriptors

Gene
Spearman · Benjamini-Hochberg · q < 0.05

Random Forest · nested

Structure
PDB 2OH4 · 2.05 Å · PyMOL

Methods

Image, gene, structure.

PyRadiomics turns each MRI into measurements we can name. Does it rise and fall together with KDR in the same patient?

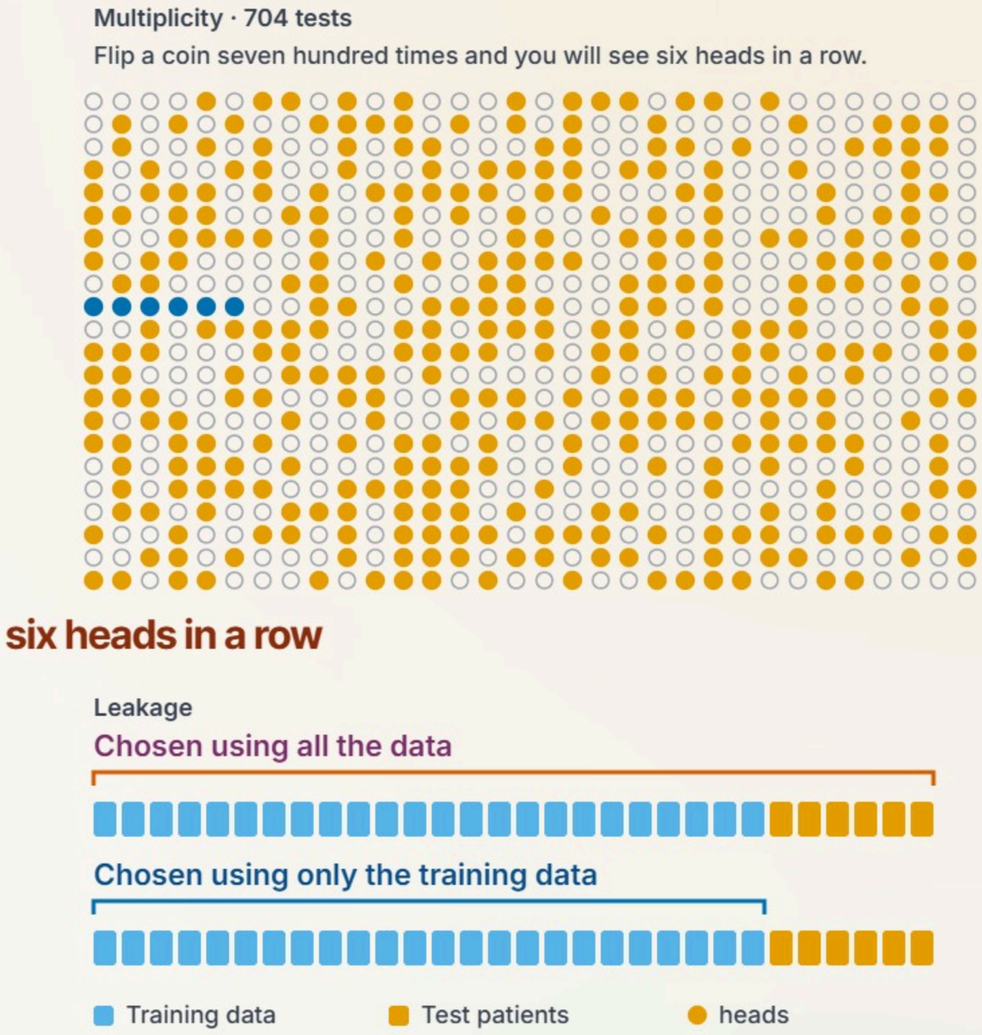
> **Why not deep learning? Ninety one patients is far too few.**
>
> And a deep learning model cannot tell a doctor which part of the image it used.

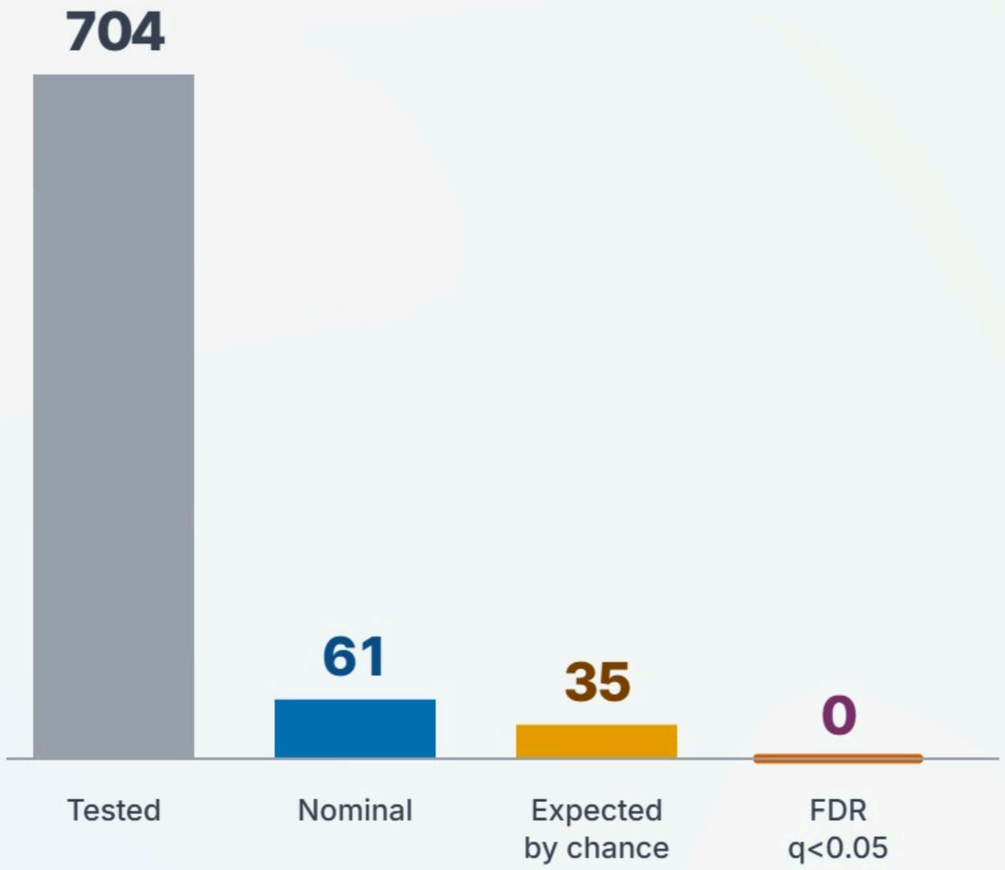
Two checks

Two checks. Neither was in my first draft.

01 **Multiplicity** 02 **Leakage**

Both are standard. And together, they are the reason this talk exists.





Check 01 · Multiplicity

Zero survived.

704
tests, one per measurement

61
came back significant

35
expected by chance alone

0
survived FDR correction

If nothing were true at all, chance alone would still give us about thirty five.

Not one out of seven hundred and four. Strongest: q = 0.530

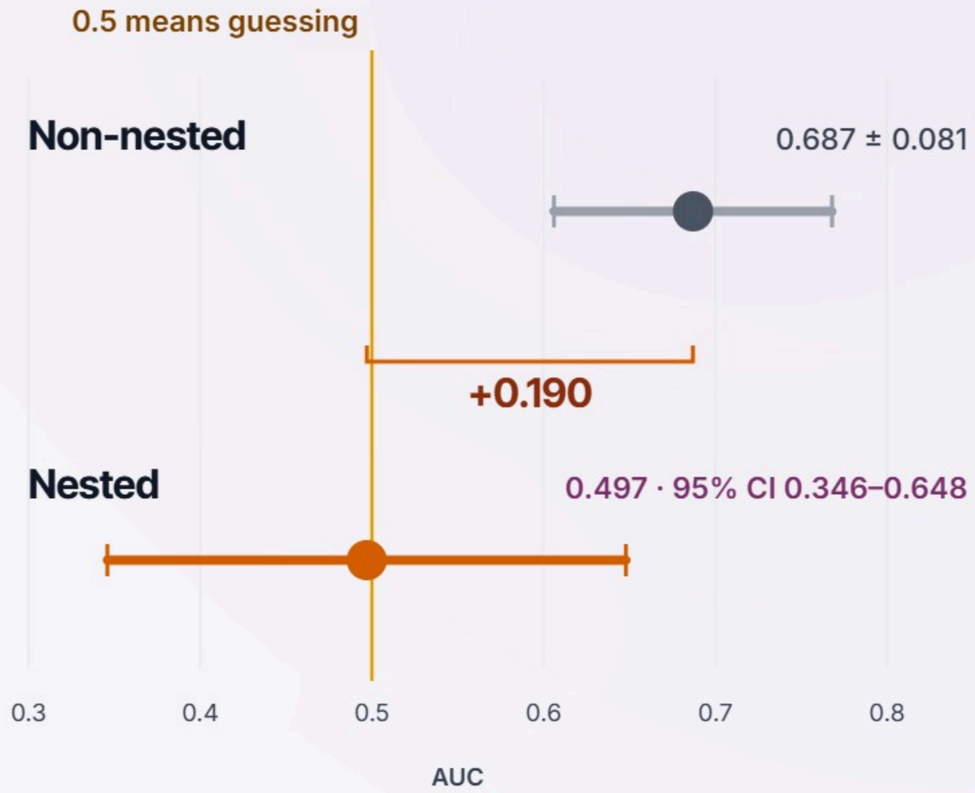
Check 02 · Leakage

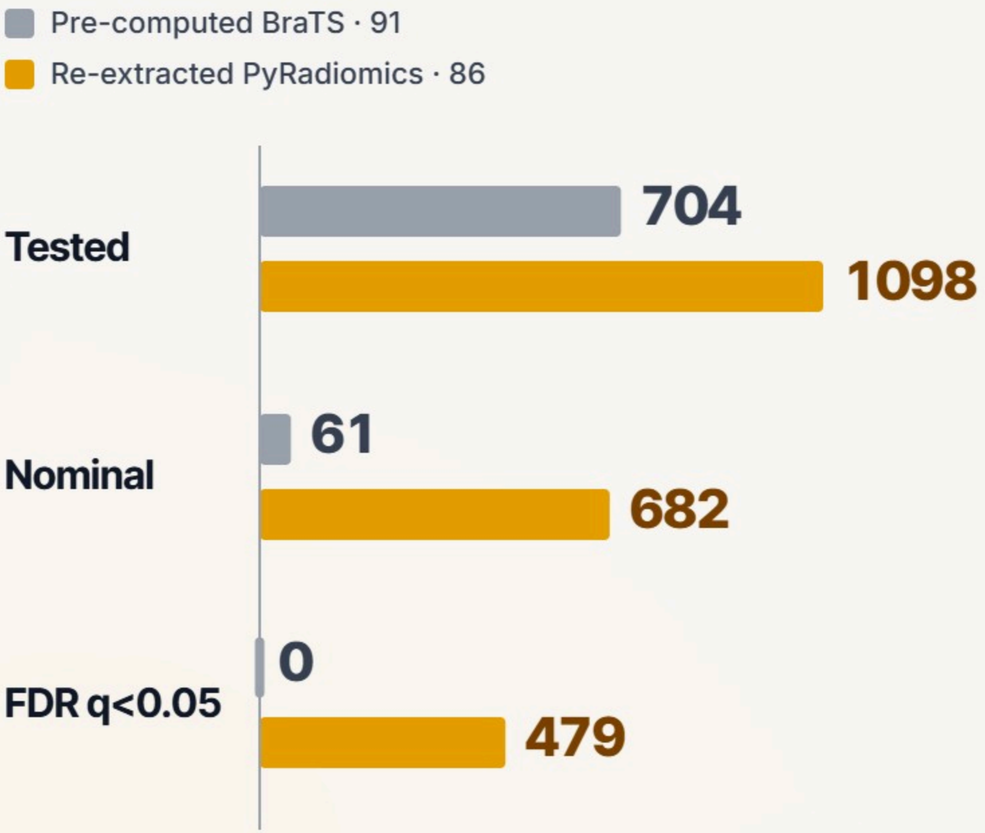
I used all the data.

The model was not reading the tumour. It was reading my own choices back to me.

Of a hundred patients, about fifty go into the wrong group.

+0.190 — all of it came from the mistake





Provenance

Same images. Different software. Opposite answers.

This fits a software effect, but it does not prove one.

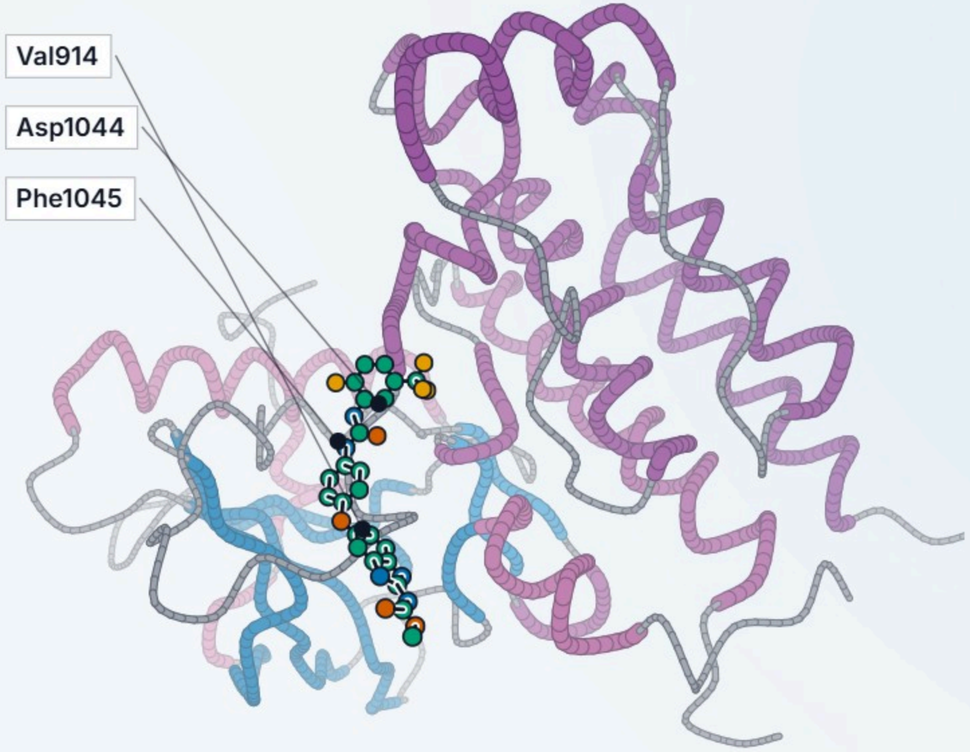
Not 479 independent findings.

Of six genes, KDR was the strongest: q = 0.010

Discussion

The biology did not change. The record keeping did.

Two ordinary choices, how the measurements were made and how the model was tested, changed the answer completely.

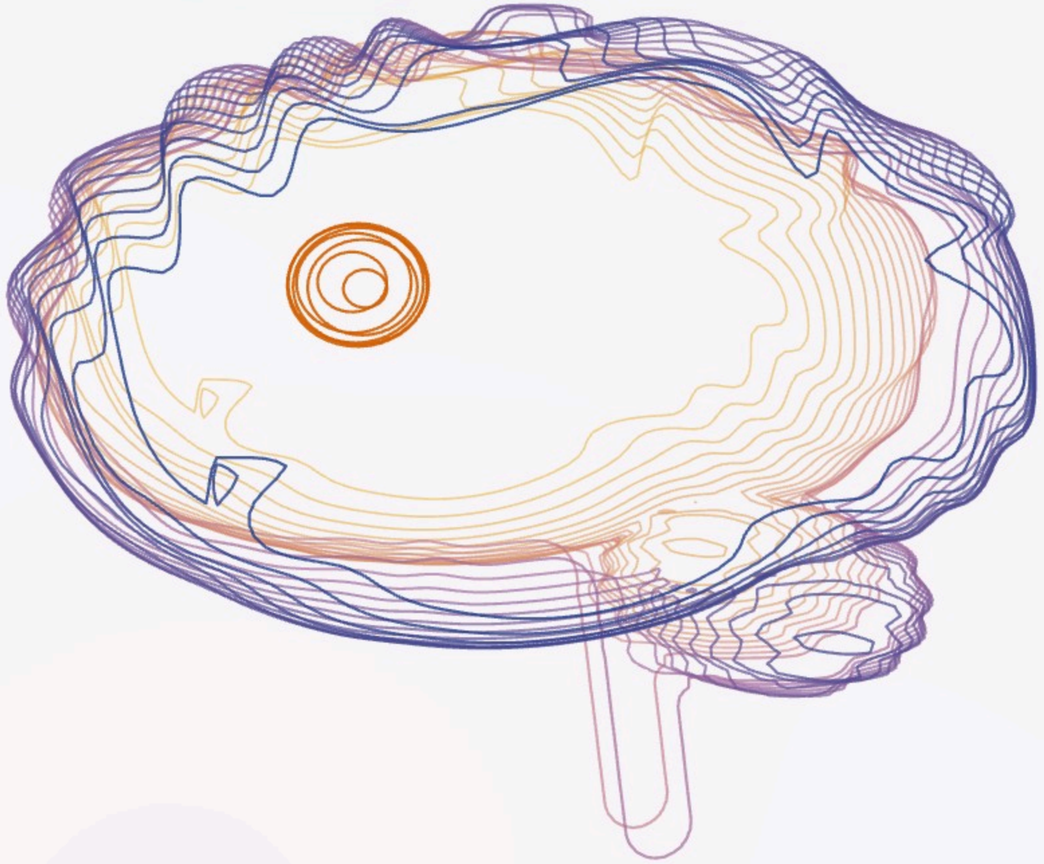


● Inhibitor ● Val914 · Asp1044 · Phe1045 — VEGFR-2 kinase domain, PDB 2OH4

Run both checks before you publish your number.

Mine took one afternoon.

Everything here is public, the code is open. I would be glad if someone tried to prove me wrong.
ipb.link/triu-nugraha-2026



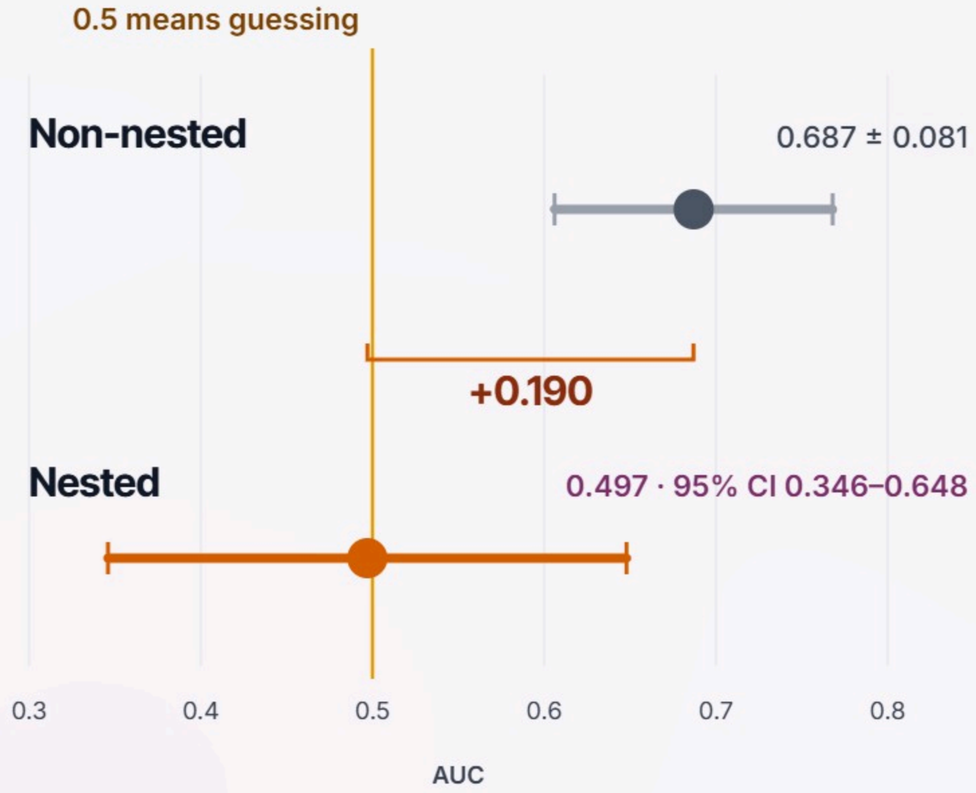
Thank you.

ขอบคุณครับ

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Extra slides: validation design · software comparison · protein structure



Backup A · Validation design

Validation design

Non-nested

Screening and tuning fitted on the full cohort.

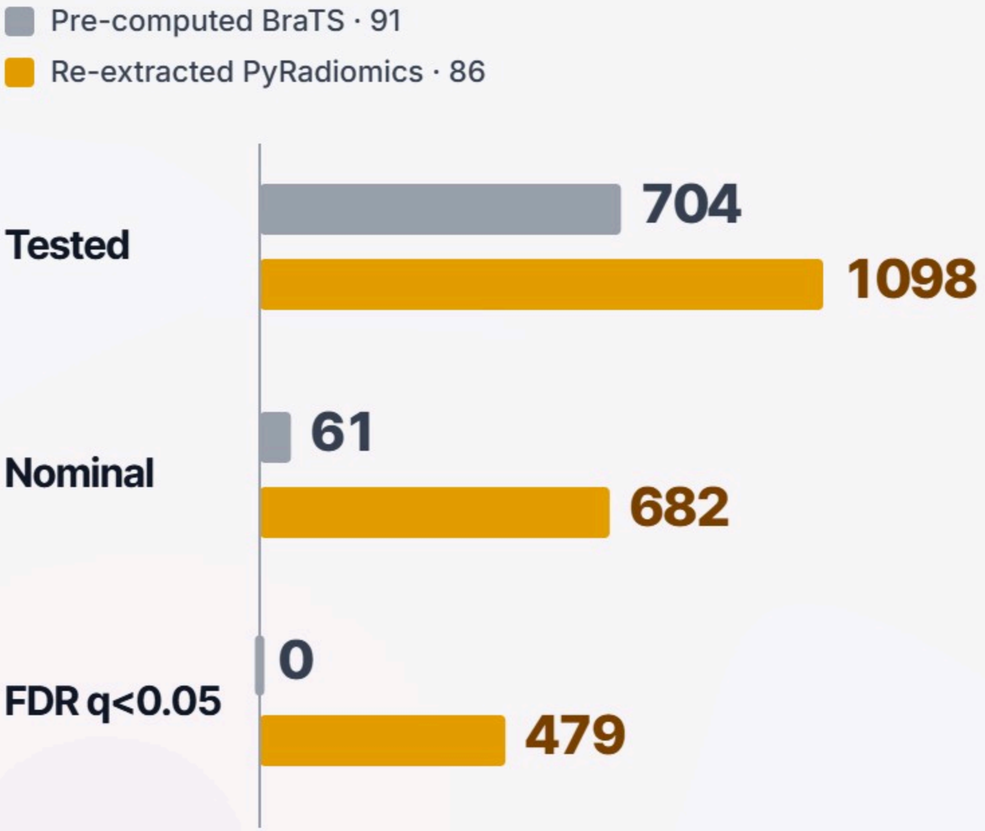
AUC 0.687 ± 0.081

Nested

Both confined to each outer training fold.

AUC 0.497 · 95% CI 0.346–0.648 · accuracy 48.4%

Median split z = −0.179 · 46 HIGH / 45 LOW



Backup B · Software comparison

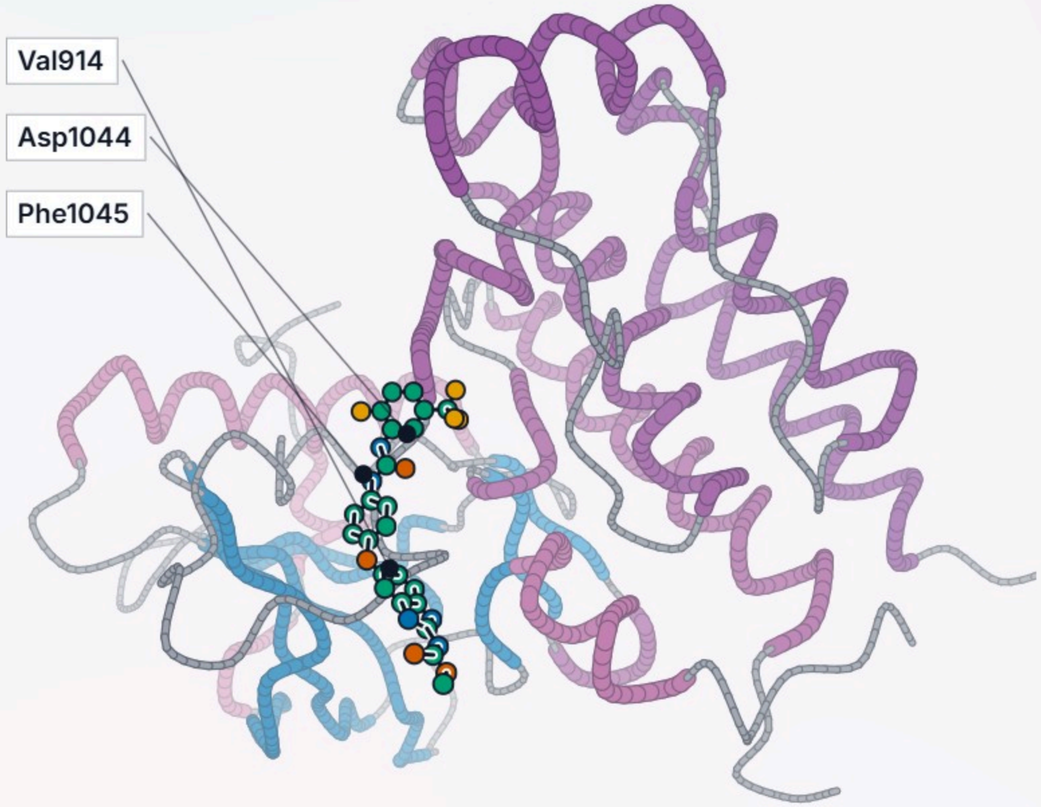
Software comparison

Table 1. Correlation with KDR expression for two descriptor sets computed from the same images.

DESCRIPTOR SET	PATIENTS	TESTED	NOMINAL	FDR Q<0.05	MIN Q
Pre-computed BraTS	91	704	61	0	0.530
Re-extracted PyRadiomics	86	1098	682	479	0.010

479 represents a few underlying dimensions expressed repeatedly, not 479 independent findings.

CXCL8 also survived correction (q = 0.039).



Inhibitor ● Val914 · Asp1044 · Phe1045 — VEGFR-2 kinase domain, PDB 2OH4

Backup C · Protein structure

VEGFR-2 kinase domain, PDB 2OH4.

01 Eight ATP-binding residues within 5 Å of the inhibitor.

02 Gatekeeper Val914

03 DFG motif Asp1044/Phe1045

04 Numbering offset by two from UniProt.

Qualitative visualisation supplying molecular context rather than a mechanistic bridge to the imaging model.